

AE 498

9/26/2026

yahmin2@illinois.edu

3. (a) Assume a 1-feature classification, and all class variances σ_i^2 are equal. ($\sigma_i^2 = \sigma^2$)

$$Pr(Y=k | X=x) = \frac{Pr(Y=k) Pr(X=x | Y=k)}{Pr(X=x)} \quad (\text{Bayes's theorem})$$

$$Pr(Y=k | X=x) = \frac{\pi_k f_k(x)}{\sum_{i=1}^C \pi_i f_i(x)} \quad \begin{array}{l} f_k(x) = Pr(X=x | Y=k) \\ \pi_k = Pr(Y=k) \\ C = \text{total no. of classes} \\ \sum_{i=1}^C \pi_i f_i(x) = Pr(X=x) \end{array}$$

If $f_k(x)$ follows a Gaussian distribution with std: σ , mean: μ_k :

$$f_k(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2\sigma^2} (x - \mu_k)^2\right)$$

$$Pr(Y=k | X=x) = \frac{\pi_k \left(\frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2\sigma^2} (x - \mu_k)^2\right) \right)}{\sum_{i=1}^C \pi_i \left(\frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2\sigma^2} (x - \mu_i)^2\right) \right)}$$

Choose k to maximize the discriminant function $\delta_k(x)$

$$\delta_k(x) = \frac{x \mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \ln(\pi_k) = \frac{2x\mu_k - \mu_k^2}{\sigma^2} + \ln(\pi_k)$$

$$(x - \mu_k)^2 = x^2 - 2\mu_k x + \mu_k^2$$

$$2x\mu_k - \mu_k^2 = -(x - \mu_k)^2 + x^2$$

$$\delta_k(x) = \underbrace{-\frac{(x - \mu_k)^2}{2\sigma^2}}_{\text{Argument of exponential in } Pr(Y=k | X=x)} + \frac{x^2}{2\sigma^2} + \ln(\pi_k) > \delta_i(x) \text{ for all categories } i$$

Argument of exponential in $Pr(Y=k | X=x)$

$$\exp(\delta_k(x)) = \pi_k \exp \left(\overbrace{-\frac{(x-\mu_k)^2}{2\sigma^2}}^{\text{rewriting as } (*)_k} \right) \exp \left(\frac{x^2}{2\sigma^2} \right)$$

$$= \pi_k \exp(*)_k \exp \left(\frac{x^2}{2\sigma^2} \right) \geq \pi_i \exp(*)_i \exp \left(\frac{x^2}{2\sigma^2} \right)$$

$$\Rightarrow \pi_k \exp(*)_k \geq \pi_i \exp(*)_i$$

Now take

$$Pr(Y=k | X=x) = \frac{\frac{\pi_k}{\sqrt{2\pi}\sigma} \exp(*)_k}{\sum_{i=1}^C \pi_i \exp(*)_i} \geq \frac{\frac{\pi_i}{\sqrt{2\pi}\sigma} \exp(*)_i}{\sum_{j=1}^C \pi_j \exp(*)_j} \quad \text{for all classes } i.$$

Numerator on the left is greater; denominators are equal on both sides.

$$\Rightarrow Pr(Y=k | X=x) \geq Pr(Y=i | X=x) \text{ for all classes } i.$$

(b) Remove the assumption of constant σ_i for all classes.

$$Pr(Y=k | X=x) = \frac{\frac{\pi_k}{\sqrt{2\pi}\sigma_k} \exp \left(-\frac{(x-\mu_k)^2}{2\sigma_k^2} \right)}{\sum_{i=1}^C \pi_i \exp \left(-\frac{(x-\mu_i)^2}{2\sigma_k^2} \right)}$$

The discriminant function must satisfy

$$k = \arg \max (\delta_k) \Rightarrow Pr(Y=k | X=x) \geq Pr(Y=i | X=x) \text{ for all classes } i.$$

To maximize $Pr(Y=k | X=x)$,

$$\frac{\pi_k}{\sqrt{2\pi}\sigma_k} \exp \left(-\frac{(x-\mu_k)^2}{2\sigma_k^2} \right) \geq \frac{\pi_i}{\sqrt{2\pi}\sigma_i} \exp \left(-\frac{(x-\mu_i)^2}{2\sigma_i^2} \right)$$

$$\ln \left(\frac{\pi_k}{\sqrt{2\pi}\sigma_k} \right) - \frac{(x-\mu_k)^2}{2\sigma_k^2} \geq \ln \left(\frac{\pi_i}{\sqrt{2\pi}\sigma_i} \right) - \frac{(x-\mu_i)^2}{2\sigma_i^2}$$

$$-\frac{(x-\mu_k)^2}{2\sigma_k^2} + \ln\left(\frac{\pi_k}{\sqrt{2\pi}\sigma_k}\right) \geq -\frac{(x-\mu_i)^2}{2\sigma_i^2} + \ln\left(\frac{\pi_i}{\sqrt{2\pi}\sigma_i}\right)$$

σ_i^2 cannot be eliminated as before, so

$$\delta_k(x) \sim -\frac{(x-\mu_k)^2}{2\sigma_k^2} + \ln\left(\frac{\pi_k}{\sqrt{2\pi}\sigma_k}\right) \quad \text{which is quadratic in } x.$$